

SOFTWARE REQUIREMENTS SPECIFICATION

EcoVision AI

Deep Learning Based Waste Classification System

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Primary Dataset	TrashNet
Project Type	Deep Learning / Computer Vision
Application	Streamlit Web Application

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1. Introduction

EcoVision AI is a Deep Learning based software system designed to classify waste images into predefined waste categories. The system uses Computer Vision and a Convolutional Neural Network based approach to analyze an uploaded image and provide a waste category prediction through a simple web interface.

This Software Requirements Specification defines the requirements, scope, users, system behavior, constraints, and technical environment of the EcoVision AI system.

2. Purpose

The purpose of EcoVision AI is to provide an easy-to-use software solution for automated waste image classification. The system is intended to demonstrate the practical use of Deep Learning for an environmental and recycling-related application.

3. Problem Statement

Manual identification and sorting of waste can be time-consuming and inconsistent. EcoVision AI aims to assist users by automatically identifying the category of a waste item from an image, providing a foundation for smarter waste-management and recycling applications.

4. Scope

The system accepts a single uploaded image and processes it using a trained Deep Learning classification model. It supports six waste categories: cardboard, glass, metal, paper, plastic, and trash.

The project includes dataset inspection, image preprocessing, model development, model saving, application development, and web deployment.

5. Objectives

- Develop a Deep Learning based image classification system for waste.
- Process and prepare image data for model training and inference.
- Classify waste images into six predefined categories.
- Provide a simple and user-friendly web interface.
- Display the predicted category and prediction confidence.
- Create a reusable trained model artifact for application deployment.
- Deploy the application through Streamlit.

6. Intended Users

- Students and learners studying Artificial Intelligence and Deep Learning.
- Users interested in demonstrating computer vision applications.
- Researchers or developers exploring automated waste classification.
- Potential future users of intelligent recycling assistance systems.

7. Functional Requirements

ID	Requirement	Description
FR-01	Image Upload	The system shall allow users to upload JPG, JPEG, or PNG images.
FR-02	Image Validation	The system shall accept only supported image formats.
FR-03	Preprocessing	The system shall resize and prepare the uploaded image for model inference.
FR-04	Classification	The system shall classify the image into one of the supported waste categories.
FR-05	Prediction Display	The system shall display the predicted waste category.
FR-06	Confidence Display	The system shall display the confidence associated with the prediction.
FR-07	Class Probabilities	The system shall provide probability information for the supported categories.
FR-08	Model Loading	The application shall load the saved trained model without retraining it.

8. Non-Functional Requirements

- Usability: The interface shall be simple and understandable for first-time users.
- Performance: The application should provide a prediction without retraining the model.
- Reliability: The application should load the saved model and process valid images consistently.
- Maintainability: Application code, model artifact, and dependencies should be kept as separate project files.
- Portability: The application should run in a compatible Python environment using the documented dependencies.
- Security: The application should process uploaded images only for the purpose of classification and should not require unnecessary personal information.
- Scalability: The system should allow future expansion to additional waste categories and larger datasets.

9. System Workflow

The proposed system workflow is:

User Uploads Image → Image Validation → Image Preprocessing → Trained Deep Learning Model → Waste Classification → Prediction and Confidence Display

10. Dataset Requirements

The primary dataset is TrashNet. The dataset contains images representing six waste categories: cardboard, glass, metal, paper, plastic, and trash.

The dataset is used for model development, validation, and testing. Images are inspected for dimensions, color format, corrupted files, duplicates, and class distribution before training.

11. Data Preprocessing Requirements

- Input images shall be converted to RGB format when required.
- Images shall be resized to the model's required input dimensions.
- Pixel values shall be normalized according to the model training pipeline.
- Training data may use suitable image augmentation to improve generalization.
- Training, validation, and testing data shall remain separated to support unbiased evaluation.

12. System Architecture

The system consists of four major components:

- User Interface: Streamlit interface for image upload and result display.
- Preprocessing Module: Converts the uploaded image into the required model input format.
- Deep Learning Model: Trained CNN/transfer-learning model used for classification.
- Prediction Module: Converts model output into a human-readable waste category and confidence information.

13. Technology Stack

- Programming Language: Python
- Deep Learning Framework: TensorFlow / Keras
- Model Architecture: MobileNetV2 based transfer learning
- Image Processing: Pillow
- Numerical Processing: NumPy
- Web Framework: Streamlit
- Development Environment: Google Colab and PyCharm
- Version Control: Git and GitHub

14. Software Requirements

- Python 3.x
- TensorFlow
- Keras
- NumPy
- Pillow
- Streamlit
- Git
- GitHub account for repository hosting
- A compatible web browser

15. Hardware Requirements

- Computer or laptop capable of running Python and a web browser.
- Adequate storage for the dataset and trained model artifact.
- Internet connection for cloud development and deployment.
- GPU-enabled environment such as Google Colab is recommended for model training.

16. User Interface Requirements

- The application shall display a clear project title.

- The application shall provide an image upload control.
- The uploaded image shall be previewed when possible.
- The predicted waste category shall be clearly displayed.
- Prediction confidence shall be visible to the user.
- The interface shall remain simple and responsive.

17. Constraints and Assumptions

- The system is limited to the waste categories represented in the training dataset.
- Prediction quality depends on the quality and visual characteristics of the uploaded image.
- The system assumes that the uploaded file contains a visible waste item.
- The deployed application requires the trained model artifact and compatible software dependencies.

18. Future Scope

- Add more waste categories.
- Use larger and more diverse real-world datasets.
- Support multiple waste objects in a single image using object detection.
- Add real-time camera-based classification.
- Fine-tune the model for improved domain-specific performance.
- Develop a mobile application version.
- Integrate the system with smart recycling or waste-management platforms.

19. Acceptance Criteria

- The application successfully starts using the documented dependencies.
- A supported image can be uploaded through the Streamlit interface.
- The uploaded image is processed without requiring model retraining.
- The system returns one supported waste category.
- The system displays prediction confidence.
- The application can be deployed using the documented project files.

20. Conclusion

EcoVision AI is specified as an end-to-end Deep Learning based waste classification system. This SRS defines what the system should do, how users interact with it, and the software and hardware environment required to operate it. Detailed model training results and evaluation metrics are intentionally excluded from this SRS and belong to the project notebook or progress/final report.